Kedros Analytics

INVENTORY OPTIMIZATION

Know What to Order, and When

Your item data already holds the answer to how much stock you need. Turn it into reorder points, order quantities and cash back on the balance sheet.

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THE OPPORTUNITY

Inventory is cash on a shelf, and most SMEs manage it by gut feel

Too much stock freezes cash, too little loses sales, and a tight cash position feels every misjudged reorder.

GUT FEEL

“We always order 100 of these.”

“Reorder when it looks low.”

This works until the business outgrows what one person can hold in their head. By then, the cost of getting it wrong has grown right along with it.

INVENTORY OPTIMIZATION

A number for every item.

Decades-old formulas replace gut feel with how much to order, when to order it, and which items deserve the closest attention. No enterprise software or data science team required.

WHAT IT IS

From “we order the usual amount” to “we know how much, and when”

Four classic formulas applied to each item’s own cost, demand and lead time.

1

Your item data

One row per SKU: unit cost, annual demand, demand variability, lead time, current stock, ordering and holding costs.

2

Run the formulas

ABC class, EOQ, safety stock, reorder point and max level are calculated for every item automatically.

3

A prioritized action list

Items at or below their reorder point are flagged, with a recommended order quantity.

Demo store: 200 synthetic SKUs · validated against business rules · results in one Excel workbook

WHAT YOU GAIN

Six ways it pays off for a small business

1 Frees up cash

Right-sized order quantities and reorder points mean less capital sitting in excess stock.

2 Reduces stockouts

Reorder points built from real demand variability and supplier lead time catch shortages before they happen.

3 Focuses attention

ABC analysis shows which items hold most of the inventory value, so scarce management time goes there first.

4 Cuts total cost

EOQ balances the cost of placing orders against the cost of holding stock, instead of optimizing only one.

5 Deliberate buffers

Safety stock is sized to a chosen service level, so the buffer is a decision rather than an accident.

6 Budgetable purchasing

Per-item order quantities turn “we might need to order soon” into a plannable cash outflow.

For a cash-constrained SME, the first two (highlighted) usually matter most.

THE FOUR NUMBERS

Four numbers replace “the usual amount”

Each one is calculated per item, from its own cost, demand and lead time.

EOQ

ECONOMIC ORDER QUANTITY

How much should we order at once?

Balances the fixed cost of placing an order against the cost of holding stock.

SS

SAFETY STOCK

How much buffer do we need?

Sized to demand variability, supplier lead time and a target service level such as 95% or 99%.

ROP

REORDER POINT

At what stock level do we reorder?

Expected demand during the lead time, plus the safety buffer.

Min/Max

OPERATING LEVELS

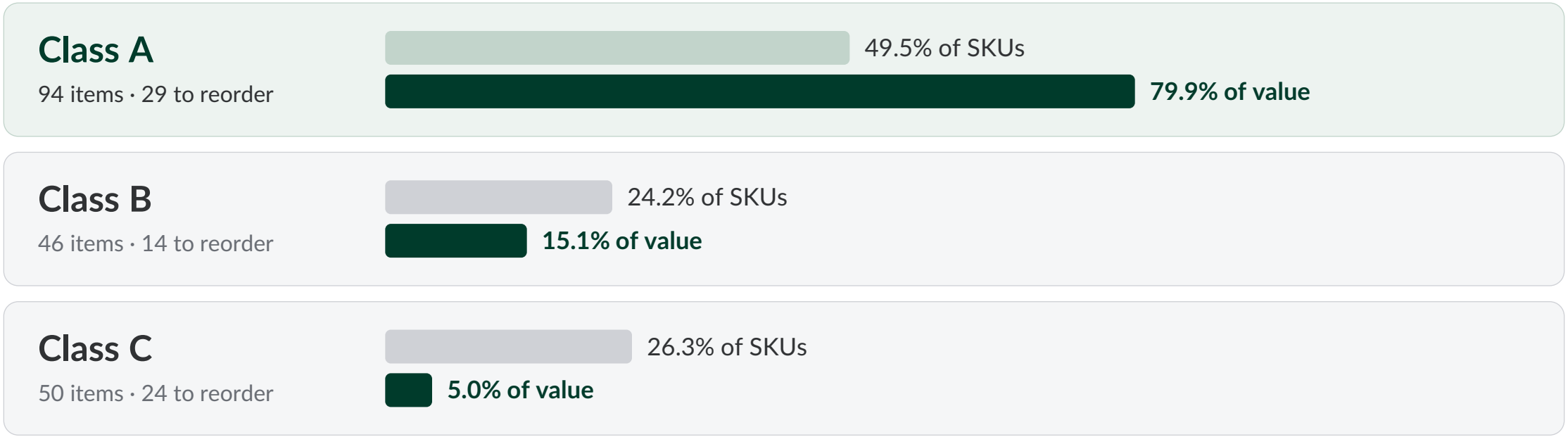
What simple rule does the warehouse follow?

Reorder at Min (the reorder point), order up to Max (reorder point plus EOQ).

EXAMPLE RESULTS

Half the items hold nearly 80% of the inventory value

Items ranked by annual consumption value (annual demand × unit cost). Demo store, 190 validated SKUs.



Synthetic demo data. Class A gets the tightest monitoring; Class C can run on simple rules.

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READING THE RESULTS

The results sheet shows a buyer what needs attention today

Two Class A items from the demo workbook: same tier, opposite situations.

SKU-00126 · ELECTRONICS

Happy Shoulder Component



Reorder now: order 29 units (EOQ)

SKU-00184 · HARDWARE

Most Animal Bundle



No action needed yet

Instead of checking every SKU by hand, the flagged rows tell a purchasing manager exactly what to order today.

Synthetic demo data. Bar lengths are scaled to units on hand.

BUSINESS IMPACT

Free up cash without cutting a single SKU

An SME with 200 SKUs and about $1M of average inventory, ordered by eyeballing stock levels.

$100,000

back on the balance sheet from a 10% cut in excess stock on $1M of inventory.

Class B and C: trim the excess

Many carry more safety stock than their demand variability and lead time justify.

Class A: protect the revenue

Tighter reorder points cut the chance of a stockout on the items that drive sales.

Illustrative example with a fictitious SME. The same effort, pointed at the items where it pays off.

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GET STARTED

What to do this week

Five steps from an item export to a reorder list.

1

Export

One row per item: unit cost, annual demand, demand variability, lead time, current stock.

2

Validate

Catch negative costs, blanks, out-of-range service levels and duplicates before analyzing.

3

Classify

Run ABC to see which items hold most of the value. Start your review there.

4

Calculate

Get EOQ, safety stock, reorder point and max level for every item.

5

Act

Review the items flagged for reorder, Class A first. Re-run as demand and lead times change.

Step 2 matters: bad rows should be caught before they ever reach the formulas.

KEEP IT HONEST

The limits worth keeping in mind

The output informs a purchasing decision. It is not the decision itself.

1 Not a full strategy

ABC, EOQ and safety stock are a foundation. A sound strategy is tailored to your decision patterns and data.

2 Needs your real numbers

Costs, demand, lead times and service levels drive every result. The demo uses synthetic data.

3 Needs clean data

Negative costs, blanks and duplicate SKUs are common in exports. Validation catches them first.

4 Reflects the past

Demand and supplier lead times drift. Re-run the analysis periodically.

HOW KEDROS ANALYTICS CAN HELP

Try it on your own data, with a free proof of concept

Business questions first, AI as a tool and not a hype machine, and results you can act on straight away.

1

Validate

Check your item export against business rules so the analysis starts from data you can trust.

2

Calculate and rank

Classify items by ABC and compute EOQ, safety stock and reorder points for each one.

3

Act and track

Work the reorder list, Class A first, and monitor cash released and stockouts over time.

Free PoC: tailored to your requirements and your company's analytics maturity.

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Ready to see what your inventory reveals?

Let’s spend thirty minutes on your item data. No commitment, no jargon, and we might even plan a free PoC.

Book a free call

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